CURRICULUM VITAE

Paul John E. Antigo

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Portfolio | LinkedIn | GitHub

Professional Profile

Fourth-year Computer Science student with practical full-stack project experience. Used AI assistance, documentation, and developer communities while working across MunLink's frontend, API, database, and deployment, implementing features, integrating application layers, and testing workflows. Contributed to undergraduate blockchain research using Solidity, TypeScript, automated testing, gas measurement, and Ethereum Sepolia. Seeking internship, OJT, and entry-level opportunities to contribute and continue learning.

Technical Competencies

Languages: Python, TypeScript, JavaScript, SQL, Solidity

Frontend: React, HTML, CSS, Tailwind CSS, Vite

Backend and APIs: Flask, REST APIs, SQLAlchemy, JWT authentication, role-based access control

Data: PostgreSQL, database design, Supabase Storage

Data analytics coursework: Microsoft Excel, SQL, Tableau, data preparation, data visualization

Blockchain and research: Hardhat, Ethereum Sepolia, Merkle trees, Keccak-256, automated testing, gas benchmarking

Engineering tools: Git, GitHub, Turborepo, Docker, Visual Studio Code, ReportLab, terminal/CLI, basic Linux CLI

Additional tools: Adobe Photoshop, Google Docs, Discord

Project Experience

MunLink | Developer and Maintainer

Civic-technology learning project | Active personal project

React, TypeScript, Tailwind CSS, Python, Flask, SQLAlchemy, PostgreSQL, Supabase Storage, JWT, ReportLab, Turborepo, Docker

Live demo

- Handled development in a three-person CSE 2 team during my third year, with teammates testing and debugging; continued maintaining and expanding the project after the course.
- Used AI tools, documentation, and community resources including Reddit to support implementation and testing, practicing detailed prompting, task decomposition, context management, output validation, and token-conscious workflows.
- Designed a multi-application architecture with separate resident and administrative React applications connected to a shared Flask REST API and PostgreSQL data model.
- Implemented resident, municipal, provincial, and super-administrator workflows scoped by province, municipality, barangay, role, and permission.
- Developed workflows for account and resident verification, document requests, payments, QR-verified PDF release, benefit programs, announcements, notifications, issue reporting, and marketplace transactions.
- Implemented JWT access and refresh tokens, password hashing, role- and location-scoped authorization, audit logging, sensitive-file access controls, and rate-limiting fundamentals.
- Performed manual functional, role-based, API, browser, database, responsive, and end-to-end workflow testing while debugging CORS, authentication, database, and frontend-to-API integration issues.
- Maintained a Zambales-scoped portfolio demo, not an official LGU deployment, with Region 3 location data retained for possible future expansion.

Research Experience

Merkle Root Framework for Micro-Credential Verification Authenticity and Gas-Efficient Issuance

Undergraduate Thesis, President Ramon Magsaysay State University | Researcher / Developer, four-member team | Final defense pending

Solidity 0.8.28, TypeScript, Hardhat, Ethereum Sepolia, Merkle trees, Keccak-256

- Contributed to the design and development of a framework that anchors one Merkle root per credential batch while preserving individual verification through Merkle inclusion proofs.
- Worked on TypeScript-based credential processing, canonical encoding, deterministic synthetic data generation, Merkle tree construction, proof generation, and proof verification.
- Participated in Solidity prototype development, issuer access control, automated testing, gas measurement, technical documentation, results analysis, and defense preparation.
- Helped execute correctness, tamper-evidence, encoding-integrity, access-control, authenticity, and cross-implementation TypeScript/Solidity verification tests.
- Participated in a controlled four-arm evaluation covering batch sizes from 1 to 3,000 credentials with three repetitions per measurement cell and live-network confirmation on Ethereum Sepolia.
- The supplied study reports approximately 48,730 gas per Merkle batch versus 47,712 gas for a single literature-baseline credential write; matched durability-tier comparisons broke even at two credentials.
- At 1,000 credentials, the four-arm study measured 99.9% lower anchoring gas than the literature baseline and 97.3% lower than its log-based baseline; all 12 Sepolia confirmation cells matched local measurements. Results are specific to the study setup.
- Deployed and source-verified the BaselineAnchor and MerkleAnchor Solidity contracts on Ethereum Sepolia.

Verified contracts: BaselineAnchor 0xE569...dEEA | MerkleAnchor 0x2f5D...deF8

Thesis adviser: Amelia E. Damian

Panel members: Dr. John Lenon E. Agatep, Nerissa L. Javier, Israel M. Cabasug

Professional Experience

IT Support Assistant - Cisco Department

Special Program for Employment of Students (SPES), Northern Zambales College, Inc. | Masinloc, Zambales | June 4-July 31, 2025 (20 days)

- Encoded and validated student information for identification-card processing and checked record consistency before production.
- Captured and prepared student photographs using Adobe Photoshop and organized records through Microsoft Excel and the school's information system.
- Supported the end-to-end ID production workflow using network-connected PCs, shared storage, and student files covering Grade 7 through college-level records.
- Performed quality checks on completed IDs and assisted with their organized release and distribution.

Education

Bachelor of Science in Computer Science

President Ramon Magsaysay State University, Iba Campus | 2023-2027 | Expected July 2027 | Fourth-year student

Senior High School - General Academic Strand

Northern Zambales College, Inc., Inhobol, Masinloc, Zambales | Completed S.Y. 2022-2023

Earned Course Certificates

Data Analytics Essentials
Cisco Networking Academy | August 29, 2026.
Badge verification

Microsoft Artificial Intelligence Course: Azure AI Fundamentals
TESDA Online Program | July 13, 2026.

Microsoft Cybersecurity Course: Security, Compliance, and Identity Fundamentals
TESDA Online Program | July 14, 2026.

Developing Designs for User Experience
TESDA Online Program | July 15, 2026.

Academic Recognition

- Dean's Lister, Bachelor of Science in Computer Science, President Ramon Magsaysay State University - First Semester, A.Y. 2023-2024.
- With Honors, Northern Zambales College, Inc. - First Quarter, S.Y. 2022-2023; academic average: 93.44%.
- With Honors, Northern Zambales College, Inc. - Grade 9, Third Quarter, S.Y. 2019-2020.
- First Runner-Up and Best in Music, Festival Dance Competition, PRMSU PE Culminating Activity - November 21, 2024.

Relevant Coursework

Software and algorithms: Data Structures and Algorithms, Object-Oriented Programming, Algorithms and Complexity, Programming Languages, Software Engineering I and II

Systems and data: Information Management, Operating Systems, Networks and Communications, Computer Architecture and Organization, Probability Theory and Statistics

Research and communication: CS Thesis Writing I, Scientific and Technical Writing

In progress: Automata Theory and Formal Languages, Information Assurance and Security, Introduction to Human-Computer Interaction, CS Thesis Writing II, 600-hour Practicum

Continuing Professional Development

- Studies cloud and generative AI fundamentals through self-directed learning.
- Continues MunLink development and creates mini-projects to apply unfamiliar technologies and concepts.

Additional Information

Languages: Filipino/Tagalog and English

Opportunity interests: Internship/OJT and entry-level software engineering, full-stack development, data, or general IT roles

Work arrangements: Open to remote, hybrid, or onsite opportunities